

Cloud Foundations



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- Definitions, Stories & Business Concerns
- Classical Enterprise, Why Cloud & Evolution of Cloud
- Service Models, Abstraction Levels (v1)

Labs

- Microsoft azure login.
- Create a virtual machine.
- Access the machine using putty.
- Setup logging and other security features.
- Download the logs.
- Analyze the logs

Cloud Computing

- Cloud computing is a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction. [**By NIST**]



Essential Characteristics of Cloud

- **On-demand self-service**

- Resources are instantly available when needed.

- **Broad network access**

- Access anytime or anywhere a connection to the Internet exists.

- **Resource pooling**

- Data center resources are pooled together optimizing quality of service.

- **Rapid elasticity**

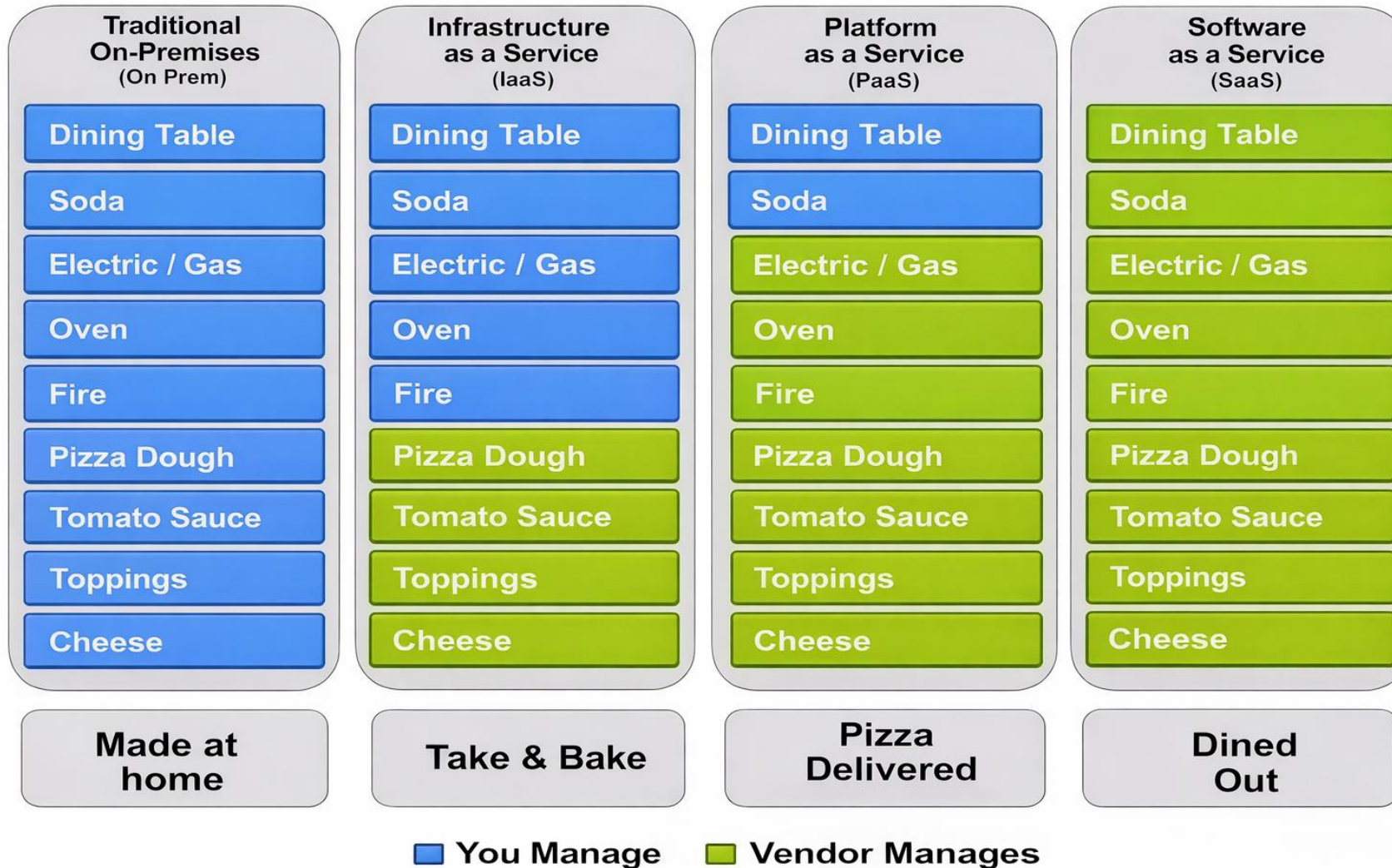
- The ability to add or remove computing resources based on need.

- **Measured service**

- The ability to measure resource usage and charge customers.

Cloud Services

Pizza as a Service



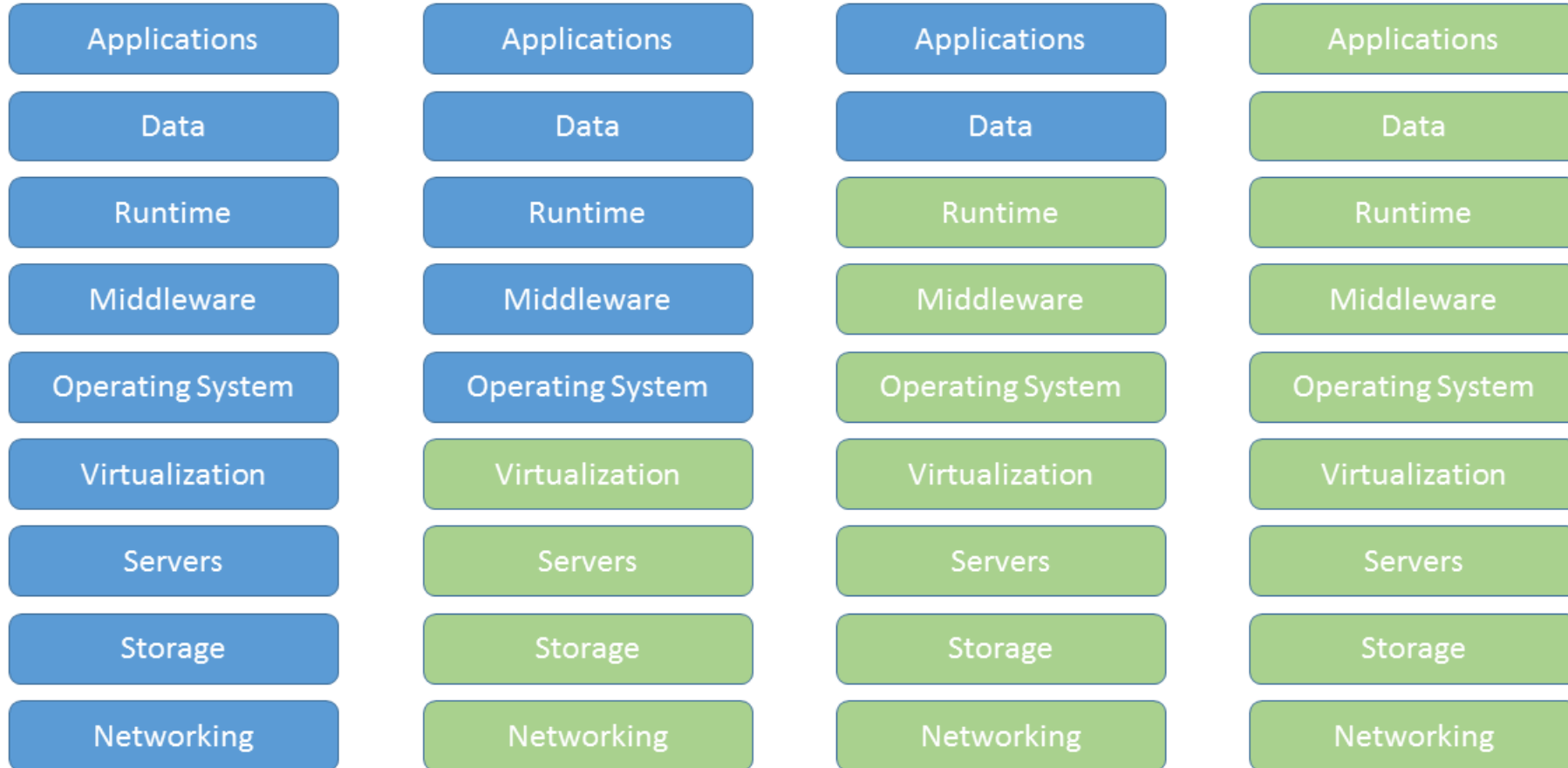
Cloud Services – Abstraction Levels

Business manages everything
(no cloud computing)

IAAS

PAAS

SAAS



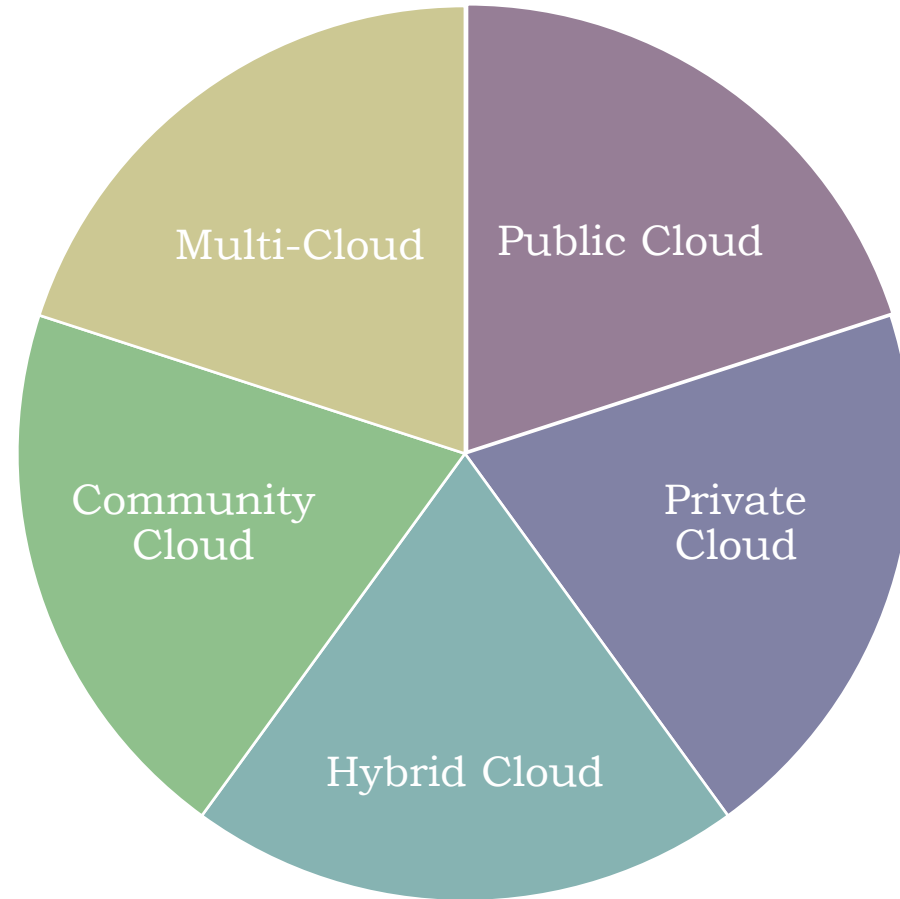
Key:

You manage

Vendor manages



Cloud Deployment Models



Business Concerns - Cost Concerns

Cost Overruns

Due to:

- Unused VMs
- Over-provisioning
- Lack of budgets/alerts

Budgeting & Predictability

- Pay-as-you-go creates variable monthly spending, unlike fixed capital expenditure in on-prem setups.



Business Concerns - Security Concerns

Misconfigurations

The #1 cause of cloud breaches:

- Public S3 buckets
- Open ports
- Weak IAM policies

Identity Attacks

- Identity is the new perimeter:
- Compromised credentials
- Stolen API keys
- Weak IAM governance

Compliance Issues

- Industries need to comply with:
- GDPR
- HIPAA
- PCI-DSS
- Failure → huge fines.

Business Concerns - Operational Concerns

Vendor Lock-in

- Moving from AWS → Azure is expensive and complex.

Downtime Impact

- Outages in major cloud providers affect business globally.

Skills Gap

- Cloud expertise shortage = reliance on consultants = high cost.

Business Concerns - Strategic Concerns

Migration Complexity

Moving large applications to cloud requires:

- Refactoring
- Re-architecting
- Re-testing

High cost & long timelines.

Performance Variability

Cloud performance may fluctuate depending on:

- Region
- VM size
- Network congestion

Business Concerns - Data Management Concerns

Data Residency

- Companies must ensure data stays within a geographical boundary (India, EU, US).

Backup & Recovery

- Businesses must manage:
- RPO (Recovery Point Objective)
- RTO (Recovery Time Objective)

Data Loss

- Improper configuration or deletion can lead to permanent loss.







That's all Folks!